CONSTRUCTIBLE SUBSETS OF AN INVERSE LIMIT OF SPECTRAL SPACES ARE PULLBACKS OF CONSTRUCTIBLE SUBSETS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of a basis of the limit topology by pullbacks of open subsets. Used inside the counted proof (source lines 5090-5100). Expresses the given constructible subset in terms of finitely many pullbacks of open sets.

Source reference. Original lemma lemma-describe-limits: sources/stacks-topology/pinned-original-source.tex, source lines 2591–2591. Navigation pointer only: the counted text refers to the Stacks item lemma-describe-limits, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{I}$ be a cofiltered category. Let $i \mapsto X_i$ be a diagram of spectral spaces such that for $a : j \to i$ in $\mathcal{I}$ the corresponding map $f_a : X_j \to X_i$ is spectral. Let $X = \lim X_i$ with projections $p_i : X \to X_i$. Let $E \subset X$ be a constructible subset. Then there exists an $i \in \mathrm{Ob}(\mathcal{I})$ and a constructible subset $E_i \subset X_i$ such that $p_i^{-1}(E_i) = E$. If E is open, resp. closed, we may choose E_i open, resp. closed.*

Proof. Assume E is a quasi-compact open of X. By Lemma lemma-describe-limits we can write $E = p_i^{-1}(U_i)$ for some i and some open $U_i \subset X_i$. Write $U_i = \bigcup U_{i,\alpha}$ as a union of quasi-compact opens. As E is quasi-compact we can find $\alpha_1, \ldots, \alpha_n$ such that $E = p_i^{-1}(U_{i,\alpha_1} \cup \ldots \cup U_{i,\alpha_n})$. Hence $E_i = U_{i,\alpha_1} \cup \ldots \cup U_{i,\alpha_n}$ works.

Assume E is a constructible closed subset. Then E^c is quasi-compact open. So $E^c = p_i^{-1}(F_i)$ for some i and quasi-compact open $F_i \subset X_i$ by the result of the previous paragraph. Then $E = p_i^{-1}(F_i^c)$ as desired.

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If E is general we can write $E = \bigcup_{l=1,\ldots,n} U_l \cap Z_l$ with U_l constructible open and Z_l constructible closed. By the result of the previous paragraphs we may write $U_l = p_{i_l}^{-1}(U_{l,i_l})$ and $Z_l = p_{j_l}^{-1}(Z_{l,j_l})$ with $U_{l,i_l} \subset X_{i_l}$ constructible open and $Z_{l,j_l} \subset X_{j_l}$ constructible closed. As $\mathcal{I}$ is cofiltered we may choose an object k of $\mathcal{I}$ and morphism $a_l : k \to i_l$ and $b_l : k \to j_l$. Then taking $E_k = \bigcup_{l=1,\ldots,n} f_{a_l}^{-1}(U_{l,i_l}) \cap f_{b_l}^{-1}(Z_{l,j_l})$ we obtain a constructible subset of X_k whose inverse image in X is E. $\square$